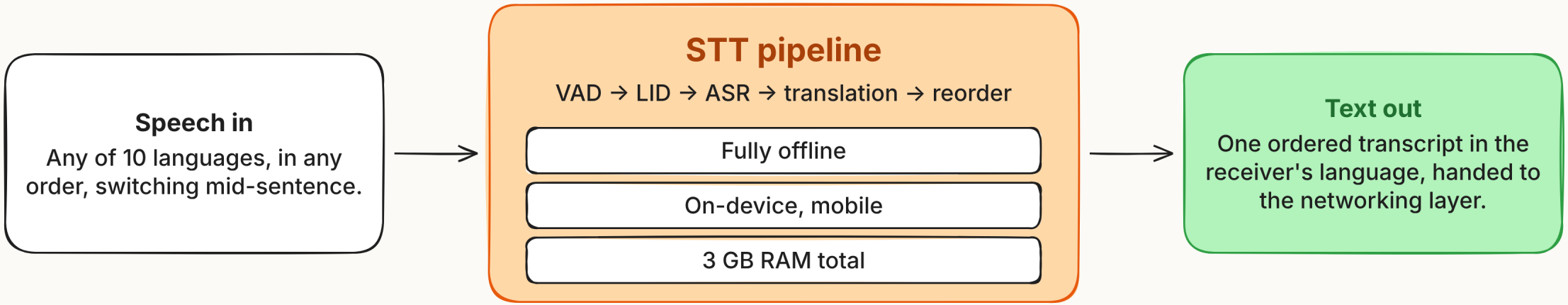
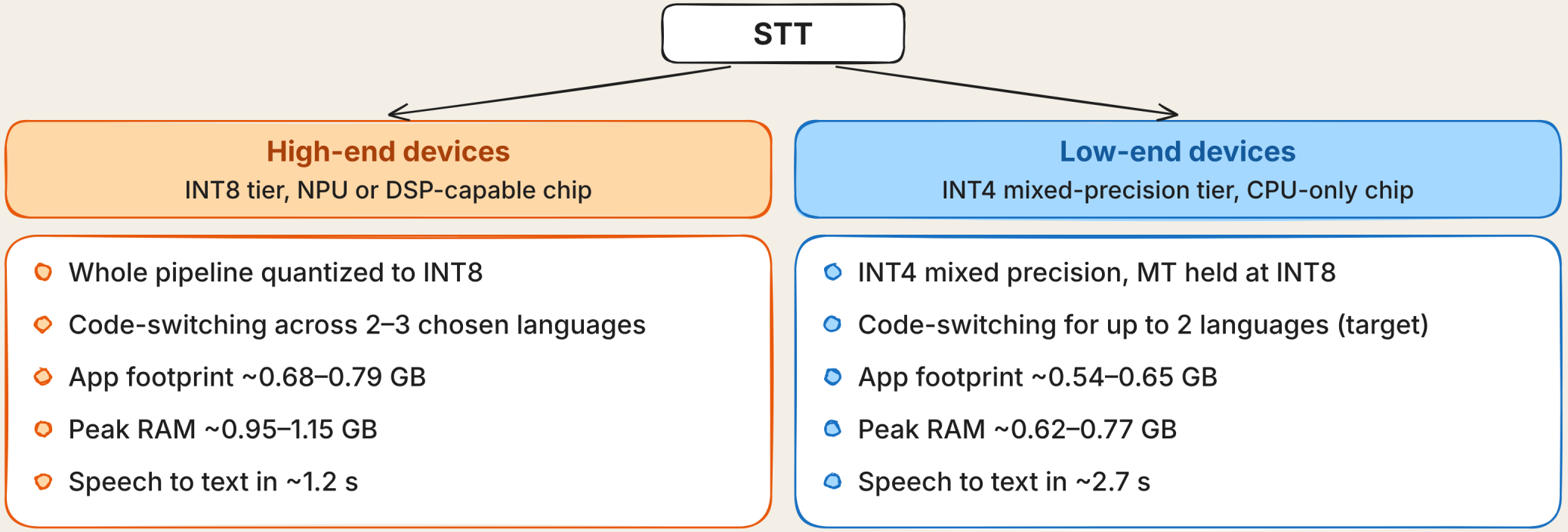
iTantra STT pipeline

Offline, on-device speech-to-text across 10 Indian languages



SCOPE

Two device tiers, one compressed pipeline



LID covers all 10 languages; each user downloads only the ones they pick

Hindi | Gujarati | Marathi | Kannada | Malayalam | Tamil | Telugu | Odia | Bengali | English

STARTING POINT · BEFORE COMPRESSION

Low-end phones began with one language, one model

Mic audio
One language for the whole session.

→

Audio to text
Vakyansh wav2vec2: one model per language.

→

Text in native language
The language that was spoken.

→

Translation
IndicTrans2, only when the receiver needs another language.

Model size per language: float32 → INT8

about 5× smaller

Hindi 378 MB → 77.3 MB	**Bengali** 378 MB → 77.2 MB	**Gujarati** 378 MB → 77.3 MB
Kannada 378 MB → 77.2 MB	**Tamil** 378 MB → ~77 MB	**Telugu** 378 MB → ~77 MB
Marathi 378 MB, INT8 unconfirmed	**Malayalam** 378 MB, INT8 unconfirmed	**Odia** 378 MB, INT8 unconfirmed

~160 MB RAM
Model footprint at run time: one model loaded.

INT8 quantization
378 MB float32 down to about 77 MB per language.

Trade-off
No code-switching: one language per session.

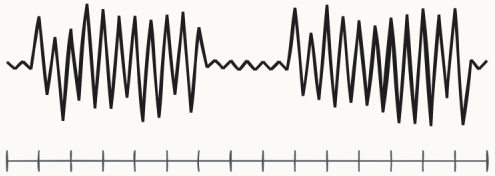
PIPELINE · STAGE 1 OF 5

VAD cuts the stream into numbered segments

Mic audio → VAD

16 kHz stream

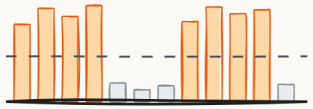
Raw audio from the mic



512-sample windows
= 32 ms at 16 kHz

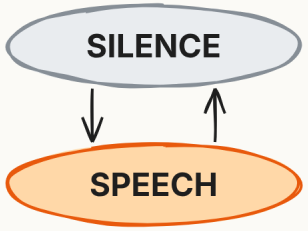
Silero VAD
conv + GRU, ~2 MB
not compressed

Speech score per window



dashed = threshold

Debounced state machine



flips only after the score holds, not on one crossing

Segment queue

- segment #3
- segment #2
- segment #1

queueing point #1

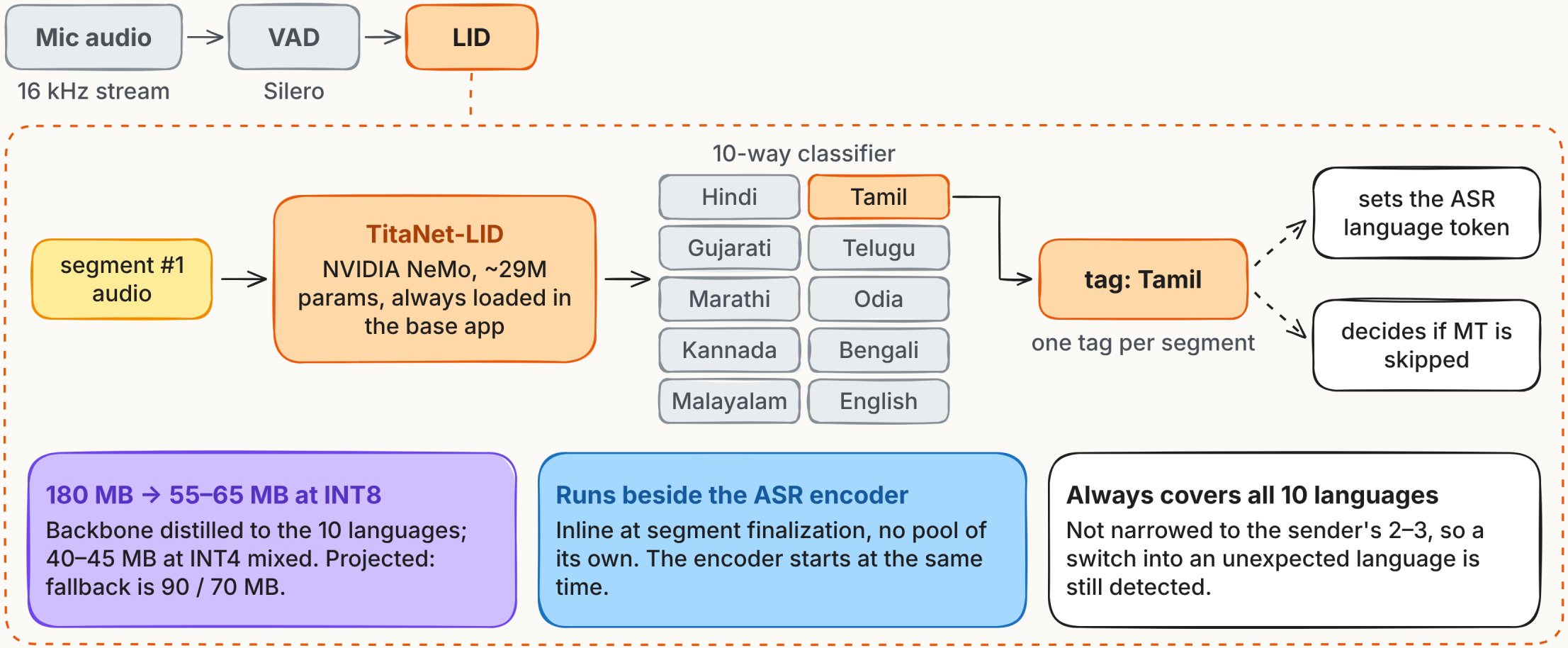
Own dedicated thread
Runs continuously and never waits for anything downstream.

Sequence number at finalization
Each finished segment is tagged, then dropped into the queue.

Known limitation
No speaker separation: overlapping speech from two people reads as one continuous region.

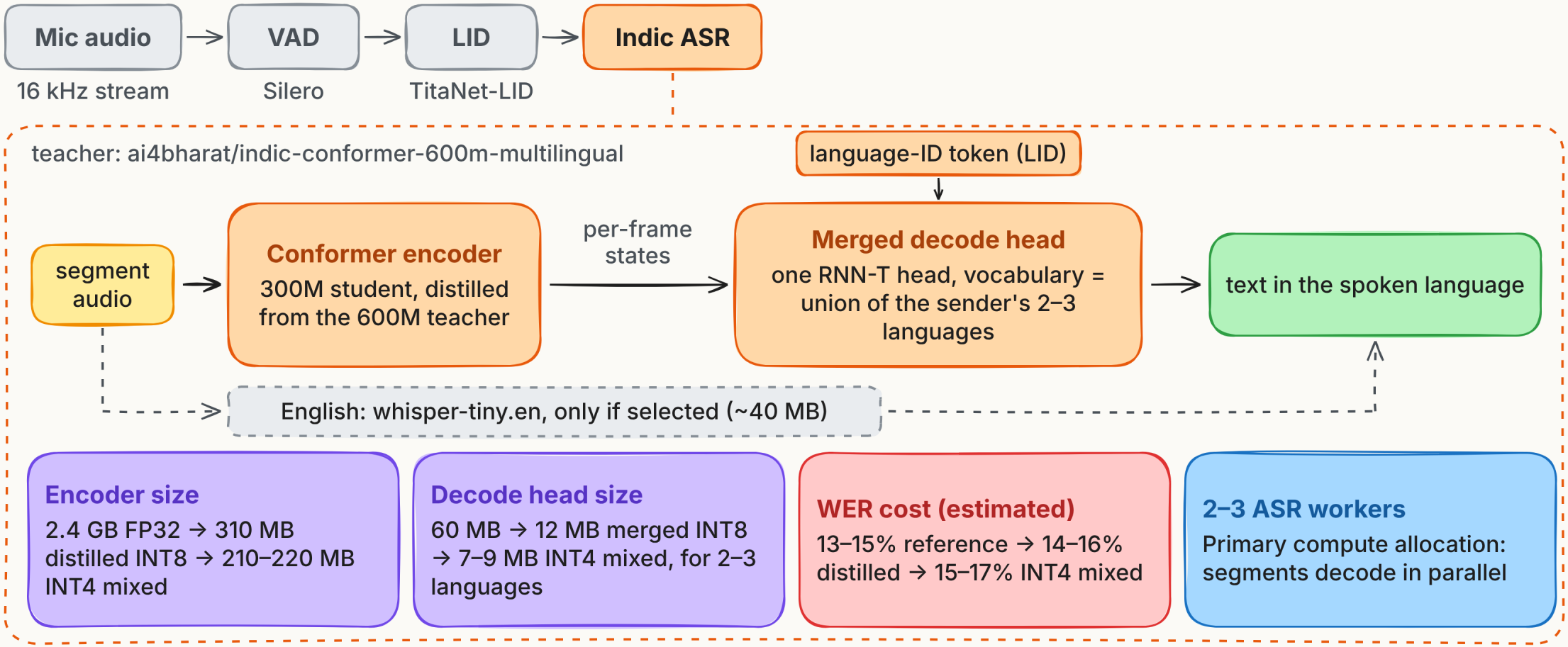
PIPELINE · STAGE 2 OF 5

LID tags each segment with its language



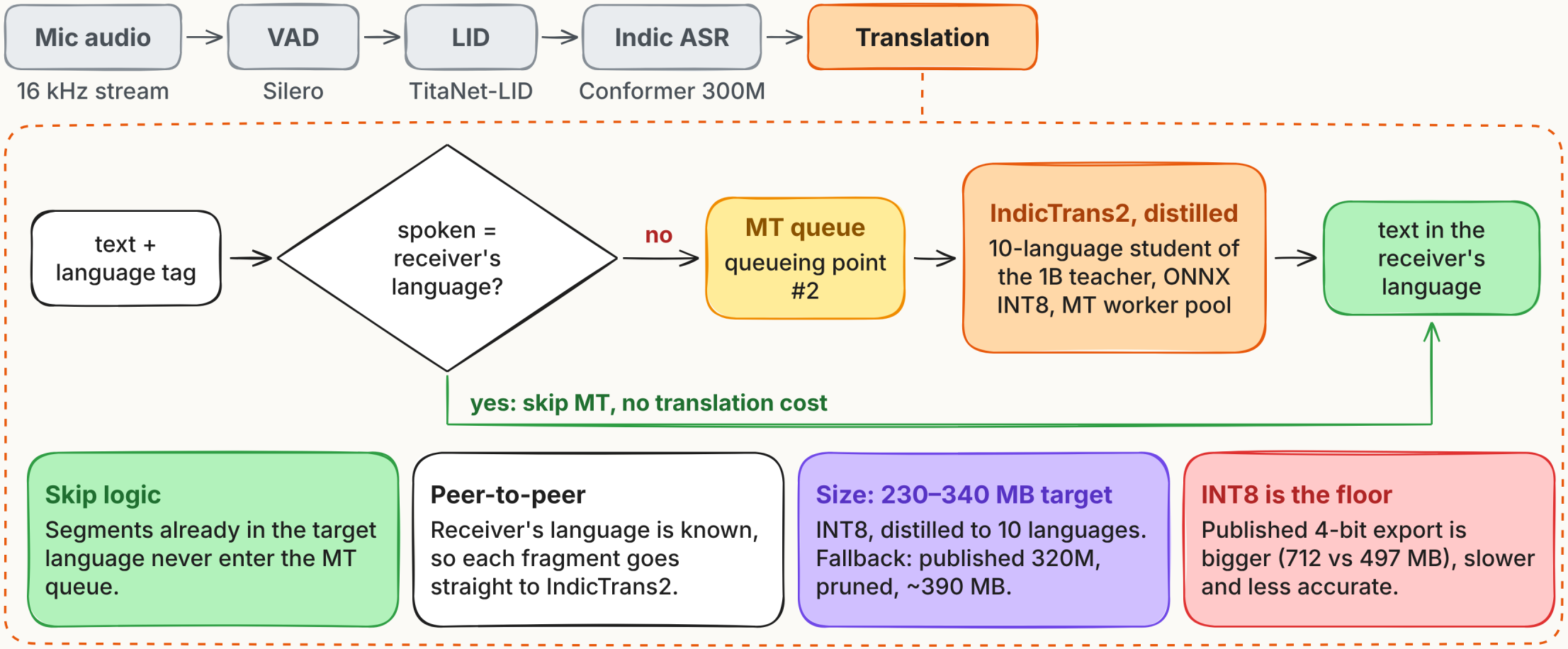
PIPELINE · STAGE 3 OF 5

A distilled encoder and one merged decode head



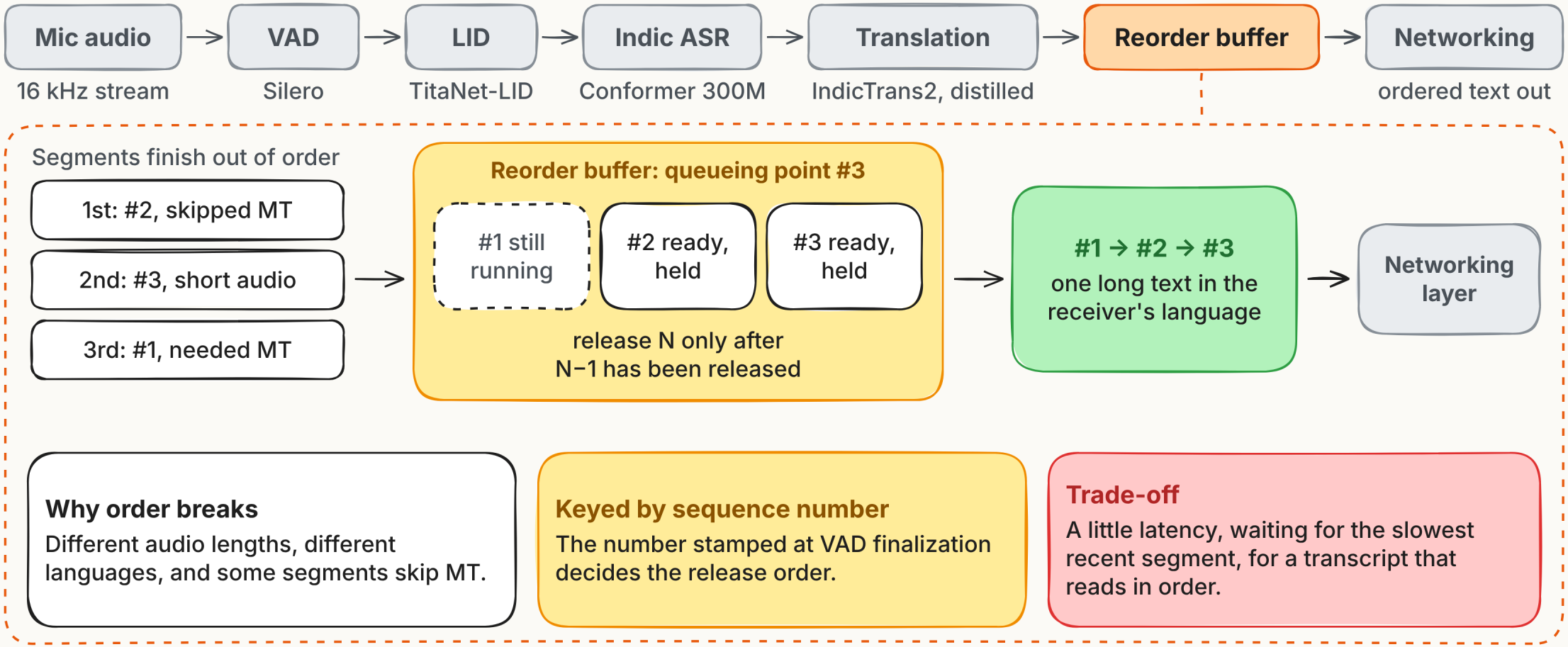
PIPELINE · STAGE 4 OF 5

Translation runs only when languages differ



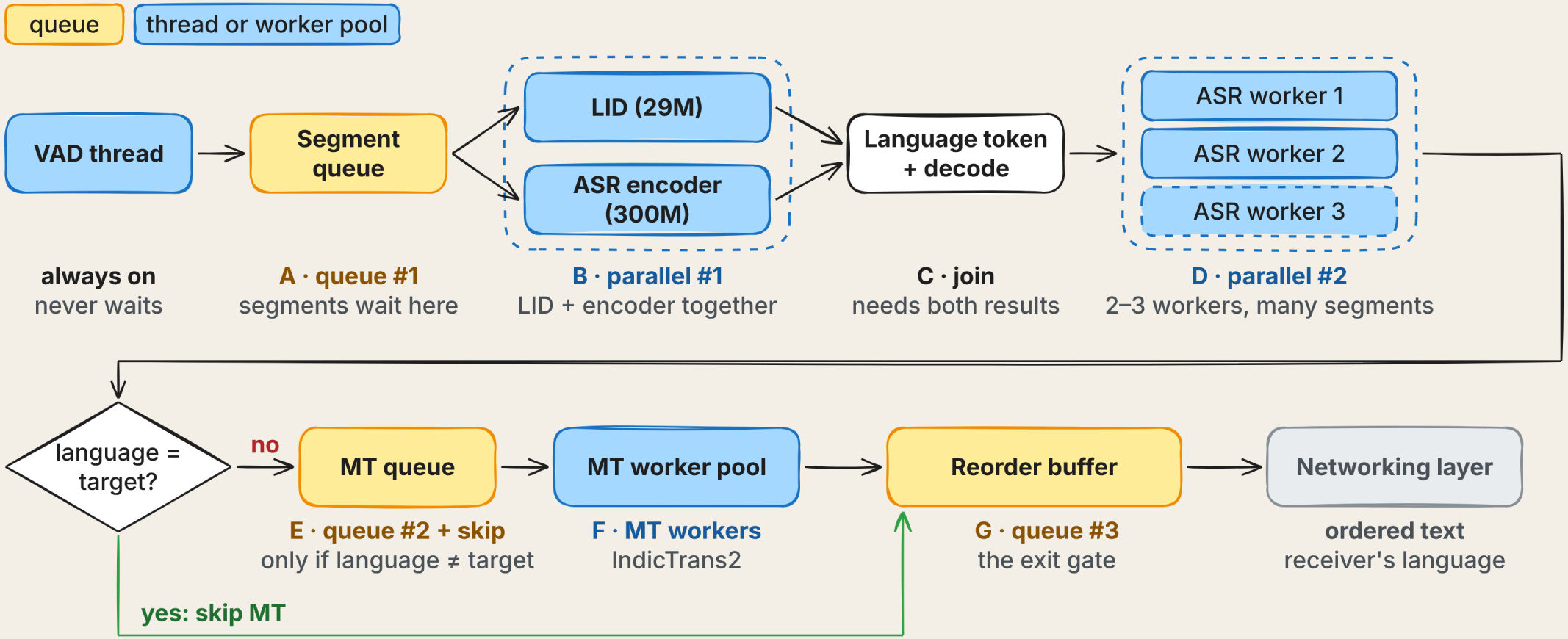
PIPELINE · STAGE 5 OF 5

The reorder buffer restores speaking order



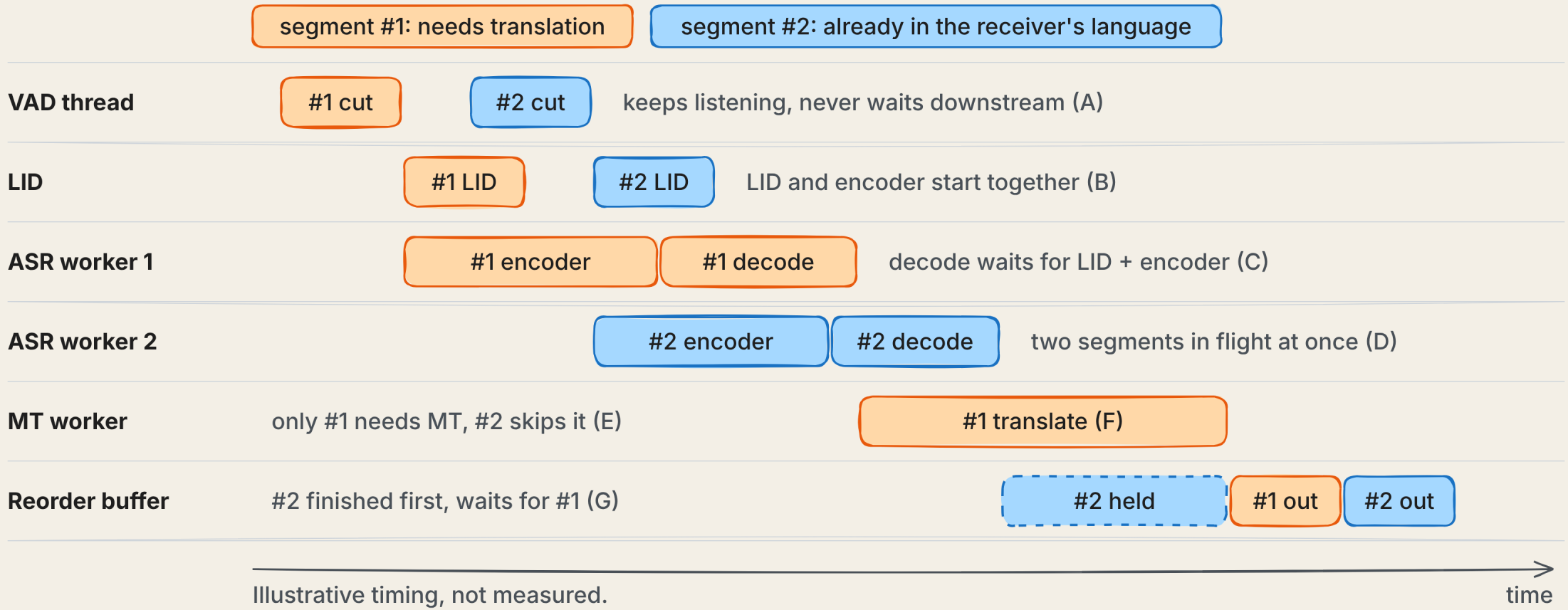
PARALLEL COMPUTING

Seven concurrency points, A to G



PARALLEL COMPUTING

Two segments moving through the threads



COMPRESSION

Seven techniques, validated model by model

1 · On-demand model packs

ASR heads, TTS voices

Base APK ships VAD, LID and the shared encoder. Other packs download for the languages the user picks.

Loads 2–3 of 9 heads

2 · Domain-restricted distillation

ASR encoder; LID and MT next

A 300M student mimics the 600M teacher, trained only on the 10 target languages.

Encoder 620 → 310 MB

3 · Vocabulary pruning

MT fallback path

Drop embedding and output rows for the IndicTrans2 languages we do not target.

MT 497 → ~390 MB

4 · Mixed-precision quantization

ASR encoder, LID

Embeddings, early layers and attention stay INT8. Mid and late FFN layers go INT4, recovered with QAT.

310 → 210–220 MB

5 · Merged decode head

ASR decode

One shared head steered by a language-ID token, instead of separate per-language heads.

Heads 15–20 → 10–13 MB

6 · Minimal ONNX Runtime

Runtime binary

ORT Mobile built with only the operators our models use.

Several MB off the app

7 · Weight clustering

Optional final pass

k-means codebook after quantization, validated per model before adoption.

About 5–10% more

Exception: MT stays at INT8

Per the published model

Its 4-bit export is larger (712 vs 497 MB), slower, and scores SacreBLEU 71.6 vs 87.1.

INT8 in both tiers

COMPRESSION

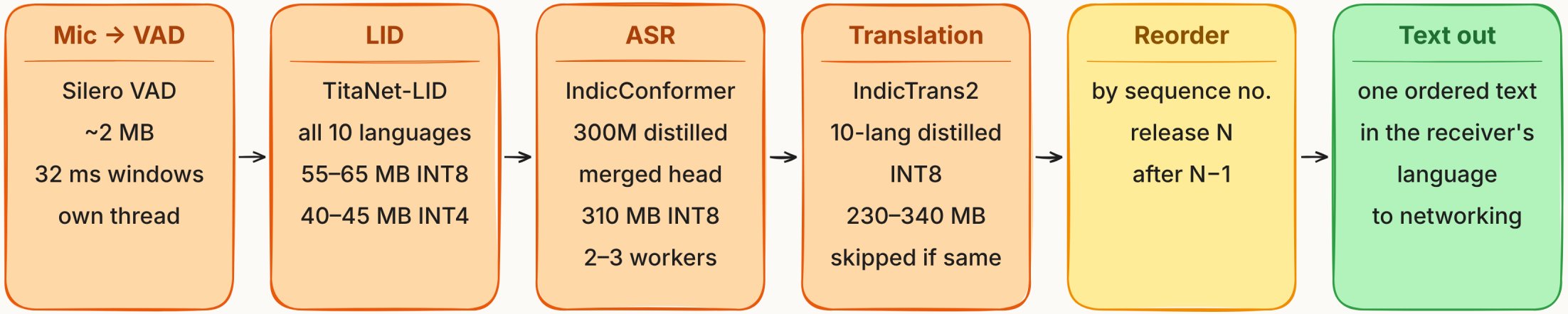
From 7.85 GB to under 0.8 GB

Component	Original, FP32	High-end, INT8	Low-end, INT4 mixed	What shrinks it
VAD	4 MB	2 MB	2 MB	Already minimal, left as is
LID	180 MB	55–65 MB	40–45 MB	Backbone distilled to 10 languages, then quantized
ASR encoder	2,400 MB	310 MB	210–220 MB	Distilled 600M → 300M, mixed precision with QAT
ASR decode heads	180 MB, 9 heads	12 MB, merged	7–9 MB, merged	On-demand packs and one merged head
Machine translation	1,870 MB	230–340 MB	230–340 MB, INT8	Distilled to 10 languages, never below INT8
Complete app, incl. 1 TTS voice	**~7.85 GB, 10 languages bundled**	**~0.68–0.79 GB**	**~0.54–0.65 GB**	**About 90% smaller**

LID and MT sizes are projections for checkpoints still to be distilled.
Validated fallbacks: LID about 90 / 70 MB, MT about 390 MB.

END TO END

From microphone to ordered transcript



Across the whole pipeline

Concurrency
Three queues (segment, MT, reorder) and two parallel points (LID with the encoder, 2–3 ASR workers).

Compression
On-demand packs, distillation, mixed-precision INT8 and INT4, a merged decode head, and the MT skip.

Design target
Fully offline on ONNX Runtime Mobile. Peak RAM about 1 GB on high-end, inside the 3 GB budget.

EVALUATION · EFFICIENCY

Footprint and resource use by device tier

Metric	High-end · INT8, NPU	Low-end · INT4 mixed, CPU
Base package, always resident (VAD + LID)	~57–67 MB	~42–47 MB
Base APK install size	~75–85 MB	~58–65 MB
App footprint, 2 STT languages + 1 TTS voice	~0.68–0.79 GB	~0.54–0.65 GB
Peak RAM, full pipeline active	~0.95–1.15 GB	~0.62–0.77 GB
RAM while idle listening	~100–120 MB	~85–100 MB
CPU while idle listening (VAD only)	~1–3% of one core	~3–6%, CPU-only
Code-switching	Full, 2–3 languages	Up to 2 languages (target)

Engineering projections, not yet measured on device. Assumes Indic-only sender languages and memory-mapped models when idle.

EVALUATION · ACCURACY

Recognition and translation quality by device tier

Metric	High-end · INT8, NPU	Low-end · INT4 mixed, CPU
STT WER, average case (22 s audio, natural pauses)	~15–18%	~17–20%
STT WER, worst case (30 s continuous speech)	~19–23%	~21–26%
LID error rate, after distillation	<1%	<1.5%
MT quality, SacreBLEU vs FP32 oracle	87.13 (placeholder)	87.13 (placeholder)
MT quality, exact text match	72.36% (placeholder)	72.36% (placeholder)

Engineering projections. MT scores are placeholders from the parent 320M checkpoint. Uncompressed reference WER: ~13–15%.

EVALUATION · LATENCY

Time to text and round trip by device tier

Metric	High-end · INT8, NPU	Low-end · INT4 mixed, CPU
ASR real-time factor (encoder + decode)	~0.12–0.18	~0.28–0.40
STT completion, average, same language	~1.2 s	~2.7 s
STT completion, average, cross-language (MT runs)	~1.45–1.55 s	~2.8–2.9 s
STT completion, worst case, same language	~1.85 s	~4.0 s
STT completion, worst case, cross-language	~2.2–2.3 s	~4.1–4.2 s
Voice round trip, average (phone A to phone B)	~2.1 s	~4.3–4.4 s
Voice round trip, worst case	~3.0–3.1 s	~6.0–6.1 s

Engineering projections. STT completion = last word spoken to text finalized. Low-end round trip before compression: 6.0–8.6 s.

EVALUATION · SUMMARY

High-end and low-end at a glance

	Total footprint	Idle CPU cost	Avg round-trip voice latency	Blended STT WER	MT quality (SacreBLEU)
High-end INT8, NPU	~0.68–0.79 GB	~1–3%	~2.1 s	~15–18%	87.13*
Low-end INT4 mixed, CPU	~0.54–0.65 GB	~3–6%	~4.3–4.4 s	~17–20%	87.13*

Before compression

~7.85 GB with all 10 languages bundled. Low-end round trip 6.0–8.6 s on average. Reference WER ~13–15%.

Status of these numbers

Engineering projections, not yet measured on device. * MT score is a placeholder from the parent 320M checkpoint.